

OLIVER CLARKE

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Education

University of Manitoba

Bachelor of Science in Computer Engineering

GPA: **4.03** (President's Scholars Society)

Sep. 2022 – Present

Anticipated Graduation: 04/28

Winnipeg, MB

Experience

DV Hardware Engineering Co-Op

Advanced Micro Devices (AMD)

May 2025 – Present

Markham, ON

- Maintain **C++** framework used to interface with SOC DFT infrastructure (JTAG, scan insertion)
- Write verification test sequences for **MI-series GPUs**, focusing on blocks in the DFX IP
- Collaborate with IP teams to define and review test plans to improve **pre-silicon** coverage
- Debug RTL and testbench failures with simulation tools and **waveform analysis**
- Support **post-silicon validation** team after tapeout by debugging failing sequences on physical hardware

Junior Software Engineer

Birchwood Automotive Group

December 2022 – February 2025

Winnipeg, MB

- Built a custom **CSS compiler** that was **10x** faster than alternative solutions, and provided ergonomic developer features such as namespacing, a preprocessor system, and composability primitives.
- Lead a major refactor from an internal PHP framework to **Laravel** for better developer productivity and familiarity
- Develop and maintain **full-stack** web applications using PHP, JavaScript, MySQL, and Docker for internal information sharing and collaborating information (ERP)
- Maintained the internal PHP web framework, fixing vulnerabilities and adding housekeeping features to make developer's lives easier
- Extensively used the **Agile** management methodology supported by Jira, Bitbucket and git

Projects

Custom Language Compiler | C, Parsing, LLVM, Semantic Analysis | [GitHub](#)

May 2025 - August 2026

- Compiler for a C-like language with modern and ergonomic features. Main focus is compile-time evaluation
- Uses **data oriented** principles for instant compilation times, minimizing developer turnaround time
- Implemented a lightweight **x86_64** backend to speed up developer iteration, compared to the **LLVM** backend
- Sophisticated dependency system, allowing order independent code and **compile time evaluation** of bytecode
- Constructed an **LSP** for smooth IDE integration to increase developer experience (DX)

Custom Text Editor | Zig, Text Editing, VIM, Data Structures | [GitHub](#)

May 2026 - Present

- Built a text editor based on VIM, using **data oriented** concepts for efficient text editing
- Implements a paged **Rope** data structure for efficient text manipulation in large files (>1 Million lines)
- Uses **DirectX** for smooth GUI rendering, and enabling custom shader effects

Wayland Compositor | Linux, DRM, Rendering

May 2025 - October 2025

- Implement a C++ display server and window manager using the **Wayland** protocol for a frame-perfect desktop
- Built everything from scratch including a wayland protocol XML scanner, the network stack using *io_uring* for asynchronous client communication, DRM swapchain rendering and presentation, etc.
- Work closely with the Linux graphics stack: DRM, EGL/GBM/MESA, and dissect GPU driver source code (AMDGPU)
- Use the Linux **DBUS** system to drive communication with desktop elements such as the taskbar, window properties, and compositor configuration

Technical Skills

Languages and Technologies: C, C++, Rust, Verilog, Valgrind, DirectX, DRM, x86-64, ARM64, Git, Atlassian Products

Operating Systems:

- Linux (have programmed simple device driver, familiar with overall kernel and user structure)
- Windows (Win32 app development, WDDM driver development)